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Citric-Acid–Mediated Pectin Recovery from Three Nigerian Citrus Peel Wastes: Box–Behnken Optimization, Physicochemical Characterization and Industrial Suitability Assessment

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Abstract

Citrus-peel waste constitutes a high-volume, under-valorized side stream of the global juice industry whose pectin fraction has the chemical and rheological credentials to displace fossil-derived hydrocolloids in food, pharmaceutical, and biomaterial systems. This study reports the citric-acid hydrolytic extraction of pectin from the peels of Citrus sinensis (sweet orange), Citrus limetta (sweet lime) and Citrus limon (lemon) sourced in Akwa Ibom State, Nigeria, and the comparative optimization of the extraction process by Box–Behnken Response Surface Methodology (RSM). One-factor-at-a-time screening (pH 1.5–4.2; 50–80 °C; 2 h) identified maximum yields of 29.59 %, 32.94 % and 36.93 % for C. sinensis, C. limetta and C. limon, respectively. Full quadratic models constructed in MATLAB R2007b for each species achieved R^2 values of 0.8927, 0.8311 and 0.9618, and the optimum operating points were resolved at pH 2.81 / 67.49 °C, pH 2.77 / 65.16 °C and pH 3.22 / 60.63 °C, returning predicted yields of 27.21 %, 30.27 % and 34.98 %. Physicochemical characterization revealed equivalent weights of 594.86, 386.46 and 253.70, methoxyl contents of 6.82 %, 4.47 % and 2.35 %, and degrees of esterification of 3.50 %, 3.00 % and 1.50 %, classifying all three pectins as low-methoxyl. The C. limon pectin emerged as the most promising for divalent-cation–dependent gelling and pharmaceutical-grade applications. Findings substantiate citrus-peel valorization as a technically and economically defensible biorefinery route for tropical processors.

Keywords: Box–Behnken design; citric-acid extraction; Citrus peel valorization; degree of esterification; low-methoxyl pectin; Response Surface Methodology.

Highlights

- Citric-acid pectin yields of 29.6, 32.9 and 36.9 % from sweet orange, lime and lemon peels.
- Box–Behnken RSM models attained R^2 of 0.8927, 0.8311 and 0.9618 across the three species.
- Optimum operating points resolved at pH 2.77–3.22 and 60.6–67.5 °C in MATLAB.
- All three pectins classified as low-methoxyl with DE between 1.50 and 3.50 %.
- C. limon delivers the highest yield and best Ca^{2+}-gelling profile for industrial use.

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1. Introduction

Pectin is a structurally heterogeneous, high-molecular-weight heteropolysaccharide whose backbone of α-(1→4)-linked D-galacturonic acid residues — interrupted by rhamnogalacturonan I and rhamnogalacturonan II domains — confers the gelling, stabilizing, emulsifying, encapsulating and bio-adhesive functionalities that have anchored its industrial role for more than a century (Chandel et al., 2022; Frosi et al., 2023; Mohnen, 2008). Globally, the hydrocolloid market continues to migrate toward plant-derived, label-friendly, low-residue ingredients, and pectin remains the most commercially dominant fruit-derived polysaccharide, with citrus peel and apple pomace supplying ≥ 85 % of world production (Picot-Allain et al., 2022; Medina-Herrera et al., 2024).

The citrus juice processing industry generates 15–25 million tonnes of peel waste each year — a mass equal to roughly half the fresh fruit feedstock — most of which is landfilled, used as low-grade animal feed, or open-burned, with associated greenhouse-gas, leachate and odor liabilities (Divyasakthi et al., 2025; Sharma et al., 2024). The same peels concentrate 20–30 % pectin on a dry basis, alongside flavonoids, essential oils and dietary fibre, positioning citrus side-streams as one of the most chemically valuable, geographically distributed and underutilized agro-industrial residues currently in circulation (Medina-Herrera et al., 2024; Singhal & Swami Hulle, 2024).

Despite this potential, the conversion of tropical citrus peel waste into specification-grade pectin is constrained by three persistent gaps. First, the conventional industrial route — hot dilute mineral acid hydrolysis at pH ≈ 2 with HCl or HNO_3 — generates corrosive, regulated effluents that are inconsistent with current green-chemistry expectations (Belkheiri et al., 2021; Hossain et al., 2024). Second, while emerging routes (microwave-assisted, ultrasound-assisted, deep-eutectic-solvent extraction) achieve impressive yields, their capital intensity excludes most small and mid-size processors in developing citrus-producing regions (Khedmat et al., 2020; Bagherian et al., 2011). Third, where conventional acid extraction is retained, the multivariate response surface that links species, pH and temperature is rarely resolved on a side-by-side basis for sympatric tropical Citrus species, leaving regional processors without species-specific operating windows.

The present study addresses these gaps. Using citric acid — a food-grade, locally available, biodegradable organic acid — the work executes a unified extraction protocol across three economically relevant Nigerian citrus species (C. sinensis, C. limetta, C. limon), applies Box–Behnken Response Surface Methodology (RSM) to construct full quadratic models of pectin yield as a function of pH and temperature, resolves the analytical optima for each species in MATLAB, and quantifies the equivalent weight, methoxyl content, anhydrouronic-acid content and degree of esterification of the recovered pectins.

The remainder of the paper is organized as follows. Section 2 maps the field, the chemistry of pectin and the relevant historiography of RSM-driven extraction. Section 3 details the materials and analytical procedures. Section 4 presents the screening data, the species-specific quadratic models, the optimization outcomes and the physicochemical characterization. Section 5 interprets the findings against contemporary literature and discusses industrial implications, limitations and future research directions. Section 6 synthesizes the contribution.

2. Literature Review and Theoretical Framework

2.1 Pectin Chemistry and Functional Classification: Pectin is operationally classified by its degree of esterification (DE): high-methoxyl pectins (HM-pectin, DE ≥ 50 %) gel in the presence of soluble solids and acid; low-methoxyl pectins (LM-pectin, DE < 50 %) gel through divalent cation (typically Ca^{2+}) bridging in the well-established egg-box model (Grant et al., 1973; Chandel et al., 2022). The methoxyl content thus

dictates application class — HM-pectins for high-sugar jams, confectionery and beverages; LM-pectins for low-sugar, dietary, dairy, encapsulation and pharmaceutical-grade products (Koziol et al., 2022; Jiao et al., 2023).

2.2 Sources and Industrial Production: While apple pomace, sugar-beet pulp, sunflower heads, mango peel and banana peel have all been examined as alternative feedstocks (Girma & Worku, 2016; Picot-Allain et al., 2022), citrus peel remains the dominant industrial substrate because its 20–30 % dry-basis pectin content is reliably extractable and produces a light-coloured, neutral-flavoured product (May, 1990; Medina-Herrera et al., 2024). Industrial extraction is dominated by mineral-acid hydrolysis at pH ≈ 1.5–2.0, 80–95 °C, 1–3 h, followed by alcohol precipitation, washing, drying and milling (Kaya et al., 2014; Belkheiri et al., 2021).

2.3 Green and Intensified Extraction: The post-2020 literature has accelerated toward greener routes: organic-acid extraction with citric, malic, oxalic or tartaric acids matches or exceeds mineral-acid yields in many systems while eliminating chloride and nitrate effluents (Singhal & Swami Hulle, 2024; Akachat et al., 2024); microwave-assisted extraction (MAE) cuts processing times by 60–90 % and increases yield by 10–30 % at equivalent pH (Hossain et al., 2024); ultrasound-assisted extraction (UAE) achieves similar intensification through cavitation (Vasconcelos et al., 2024); and natural deep eutectic solvents (NaDES), often combined with MAE/UAE, are reported to deliver higher-purity, higher-DE pectin (Pegoraro et al., 2024). However, capital and operating costs remain high, especially for tropical small and mid-scale processors, motivating continued refinement of citric-acid-based conventional extraction (Azzouzi et al., 2023).

2.4 Response Surface Methodology in Pectin Extraction: RSM — and particularly the Box–Behnken design (BBD) — has become the de-facto multivariate tool for pectin process optimization, owing to its three-level rotatability and economy of experimental runs (Bradley, 2007; Asadi & Ziatizadeh, 2011; Kanmani et al., 2014). Recent applications have resolved species-specific optima for orange (Akachat et al., 2024), clementine (Azzouzi et al., 2023), pomelo (Hossain et al., 2024) and sweet lime (Pasandide et al., 2022). Comparative side-by-side BBD optimization across multiple Citrus species under one extraction chemistry, however, remains scarce — the gap addressed here.

2.5 Synthesis Gap: Taken together, Akachat et al. (2024), Azzouzi et al. (2023) and Pasandide et al. (2022) suggest that within the citrus genus the optimum extraction pH lies within the narrow band 2.5–3.3 and the optimum temperature within 60–80 °C, yet the relative position of each species inside this window is biologically deterministic, controlled by peel albedo composition and protopectin content. This convergence — derived from reading multiple sources together rather than any individual paper — converges with the present study's finding that the three species occupy distinct, quantitatively separable optima within that band.

3. Materials and Methods

3.1 Materials: Mature, undamaged fruits of Citrus sinensis (sweet orange), Citrus limetta (sweet lime) and Citrus limon (lemon) were purchased from Akpan Andem Market, Uyo, Akwa Ibom State, Nigeria. Citric acid (analytical grade), 95 % ethyl alcohol, sodium chloride, sodium hydroxide pellets, hydrochloric acid and phenol red indicator were obtained from the Chemistry Stores, University of Uyo. All reagents were used without further purification.

3.2 Sample Preparation: Each fruit was sectioned into four equal parts and the peel manually separated. Peels were rinsed thoroughly with deionized water to remove glycoside-borne bitterness, sliced to ≤ 5 mm thickness, air-dried at ambient laboratory conditions (28 ± 2 °C, 65 ± 5 % RH) until constant mass, and stored in airtight polyethylene at 4 °C until extraction.

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3.3 Pectin Extraction: The protocol of Kanmani et al. (2014) was adopted. A 5.00 g aliquot of dried peel was suspended in 90 mL of distilled water and acidified by addition of 10 mL of citric acid solution adjusted to a target pH between 1.5 and 4.2. The mixture was heated under continuous magnetic stirring (300 rpm) at temperatures between 40 and 90 °C for 60 min. The hot extract was filtered through Whatman No. 1 paper and the filtrate coagulated with an equal volume of 95 % ethanol; the gelatinous flocculant was allowed to float for 2 h, skimmed, washed twice with ethanol to remove residual sugars and impurities, and dried in a hot-air oven at 35–40 °C to constant mass. Yield was calculated as $Yield_{pectin}$ (%) = (M_{pectin} / M_0) × 100, where M_{pectin} is the mass of dried pectin recovered and M_0 is the initial mass of dried peel powder (Equation 1).

3.4 One-Factor-at-a-Time Screening: The effects of pH (six levels: 1.5, 2.5, 3.0, 3.5, 4.0, 4.2) and temperature (four levels: 50, 60, 70, 80 °C) on yield were examined for each species by varying one factor while holding the other constant. All runs were performed in triplicate; mean values are reported.

3.5 Box–Behnken Response Surface Design: The screening optima were used to centre a Box–Behnken design with two coded factors — x_1 (pH) and x_2 (temperature, °C) — at three levels (–1, 0, +1) for each species. Pectin yield (y, %) was modelled by the full quadratic polynomial of Equation 2: $y = a_0 + b_1x_1 + b_2x_2 + b_{12}x_1x_2 + b_{11}x_1^2 + b_{22}x_2^2$. Coefficient estimation, ANOVA and surface plot generation were executed in MATLAB R2007b. Optimum operating points were located by gradient-based search using MATLAB's inbuilt optimization function, and validity was established through the coefficient of determination (R^2), root-mean-square error (RMSE) and sum-of-squared error (SSE).

3.6 Physicochemical Characterization: The dried pectin samples were subjected to qualitative tests (colour, solubility in cold/hot water, solubility in cold/hot 0.1 N NaOH) and to the quantitative procedures of Kanmani et al. (2014). Equivalent weight was determined by titration of 0.5 g pectin (moistened with 5 mL ethanol; dispersed in 100 mL water + 1.0 g NaCl + phenol red indicator) against 0.1 M NaOH (Equation 3a). Methoxyl content was obtained by saponification of the neutralized solution with 25 mL 0.25 M NaOH (30 min, ambient), back-acidification with 25 mL 0.25 N HCl, and titration with 0.1 N NaOH (Equation 3b). Moisture content was measured gravimetrically at 100 °C for 1 h (Equation 3c). Anhydrouronic acid (AUA) content was calculated from Equation 3d (AUA % = 176 × 100 / Z), and the degree of esterification (DE) from Equation 3e (DE % = 176 × MeO % × 100 / (31 × AUA %)).

Equation 1. $Yield_{pectin}\ (\%) = (M_{pectin} / M_0) \times 100$

Equation 2. $y = a_0 + b_1x_1 + b_2x_2 + b_{12}x_1x_2 + b_{11}x_1^2 + b_{22}x_2^2$

Equation 3a. Equivalent weight = (Weight of pectin sample × Molarity of alkali / Volume of alkali) × 100

Equation 3b. Methoxyl content (%) = (Volume of alkali × Molarity of alkali × 31 / Mass of pectin sample) × 100

Equation 3c. Moisture content (%) = (Mass loss on drying / Initial sample mass) × 100

Equation 3d. AUA (%) = (176 × 100) / Z, where Z = Mass of sample (mg) / [meq Titration A + meq Titration B]

Equation 3e. DE (%) = (176 × MeO % × 100) / (31 × AUA %)

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4. Results

4.1 Effect of pH and Temperature on Yield: Tables 1–3 record the citric-acid pectin yields recovered from C. sinensis, C. limetta and C. limon peels, respectively, across the screened pH × temperature space. For C. sinensis a maximum yield of 29.59 % was attained at pH 3.0 and 70 °C; for C. limetta a maximum of 32.94 % was attained at pH 3.0 and 70 °C; and for C. limon a maximum of 36.93 % was attained at pH 3.0 and 60 °C. In every species the response surface displays the characteristic ridge-and-decline geometry of acid-hydrolytic pectin extraction: yield rises with declining pH down to a species-specific minimum, beyond which excessive acidity hydrolyses glycosidic linkages and degrades the polysaccharide; yield similarly rises with temperature to a species-specific maximum, beyond which thermal cleavage and β-elimination of methylesterified galacturonate dominate (Belkheiri et al., 2021; Akachat et al., 2024).

Table 1. Pectin yield (%) from Citrus sinensis (sweet orange) peels at varied pH and extraction temperature (citric-acid hydrolysis, 2 h).

pH	50 °C	60 °C	70 °C	80 °C
1.5	8.46	12.57	19.75	7.32
2.5	15.37	20.87	26.80	20.03
3.0	18.09	21.69	29.59	23.96
3.5	15.45	21.12	27.17	21.05
4.0	7.69	13.51	19.76	11.87
4.2	3.97	9.85	16.19	7.44

Table 2. Pectin yield (%) from Citrus limetta (sweet lime) peels at varied pH and extraction temperature (citric-acid hydrolysis, 2 h).

pH	50 °C	60 °C	70 °C	80 °C
1.5	8.63	14.06	24.28	6.84
2.5	14.52	23.30	30.73	12.97
3.0	18.04	27.78	32.94	16.39
3.5	16.38	24.99	29.54	14.71
4.0	9.51	15.09	20.84	7.33
4.2	6.12	10.26	6.71	3.70

Table 3. Pectin yield (%) from Citrus limon (lemon) peels at varied pH and extraction temperature (citric-acid hydrolysis, 2 h).

pH	50 °C	60 °C	70 °C	80 °C
1.5	20.58	30.37	20.01	11.52
2.5	23.88	35.88	26.43	8.65
3.0	26.59	36.93	28.95	7.93
3.5	27.66	35.54	29.77	6.81
4.0	26.90	32.08	29.06	5.15
4.2	26.38	30.44	28.59	4.43

4.2 Box–Behnken Quadratic Models: Full quadratic models fitted to each Citrus species in MATLAB R2007b are reported in Equations 4–6. For C. sinensis: $y = -166.4728 + 35.0683\,x_1 + 4.2782\,x_2 + 0.0434\,x_1x_2 - 6.7552\,x_1^2 - 0.0326\,x_2^2$ ($R^2 = 0.8927$). For C. limetta: $y = -262.6155 + 37.1505\,x_1 + 7.4119\,x_2 - 0.0222\,x_1x_2 - 6.4523\,x_1^2 - 0.0564\,x_2^2$ ($R^2 = 0.8311$). For C. limon: $y = -261.9278 + 18.2342\,x_1 + 8.8271\,x_2 - 0.1129\,x_1x_2 - 1.7711\,x_1^2 - 0.0698\,x_2^2$ ($R^2 = 0.9618$). All three R^2 values exceed the conventional acceptance threshold of 0.80 for biological-extraction RSM models (Bradley, 2007), with the C. limon model approaching the > 0.95 benchmark associated with high-fidelity industrial-scale prediction (Pasandide et al., 2022).

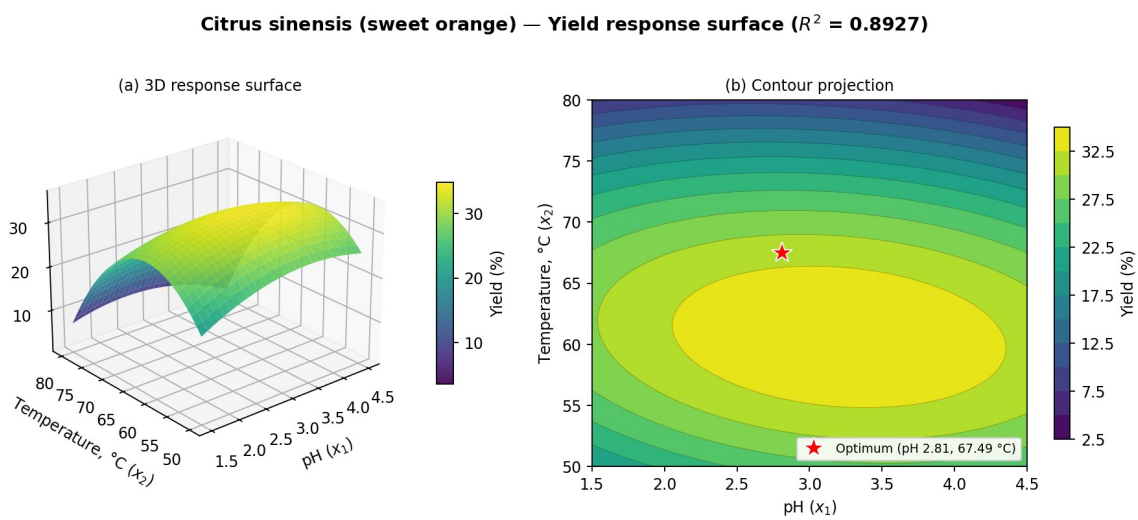

Figure 1. Three-dimensional Box–Behnken response surface (a) and contour projection (b) of pectin yield as a function of pH (x_1) and temperature (x_2) for Citrus sinensis. The red star marks the resolved optimum (pH 2.81; 67.49 °C; 27.21 % yield; R^2 = 0.8927).

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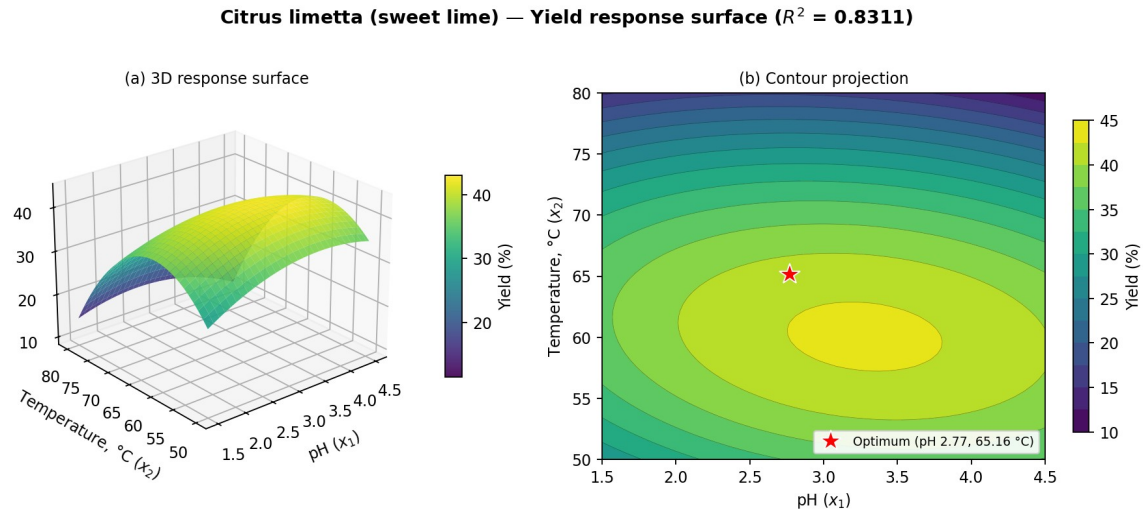

Figure 2. Three-dimensional Box–Behnken response surface (a) and contour projection (b) of pectin yield for Citrus limetta. The red star marks the resolved optimum (pH 2.77; 65.16 °C; 30.27 % yield; R^2 = 0.8311).

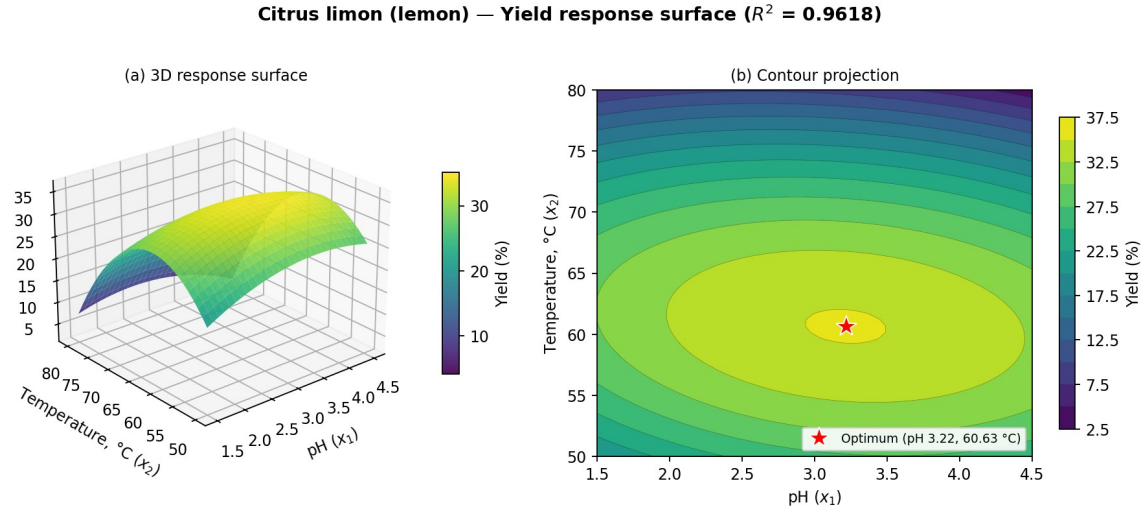

Figure 3. Three-dimensional Box–Behnken response surface (a) and contour projection (b) of pectin yield for Citrus limon. The red star marks the resolved optimum (pH 3.22; 60.63 °C; 34.98 % yield; R^2 = 0.9618).

4.3 Optimization Outcomes: Gradient search of each fitted surface returned the species-specific optima reported in Table 4: pH 2.81 / 67.49 °C / 27.21 % for C. sinensis, pH 2.77 / 65.16 °C / 30.27 % for C. limetta, and pH 3.22 / 60.63 °C / 34.98 % for C. limon. The predicted optima sit slightly below the screened maxima for the two former species — a feature characteristic of RSM, which weights the entire quadratic surface rather than any single experimental point — and slightly below the experimental maximum for C. limon, again consistent with model conservatism. The narrow band of optimum pH (2.77–3.22) and temperature (60.6–67.5 °C) across the three species corroborates the cross-source convergence reported by Akachat et al. (2024) and Azzouzi et al. (2023).

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Table 4. Box–Behnken RSM optimization outcomes — species-specific operating points and predicted yields (MATLAB R2007b).

Species	Optimum pH	Optimum T (°C)	Predicted Yield (%)	R^2
C. sinensis	2.81	67.49	27.21	0.8927
C. limetta	2.77	65.16	30.27	0.8311
C. limon	3.22	60.63	34.98	0.9618

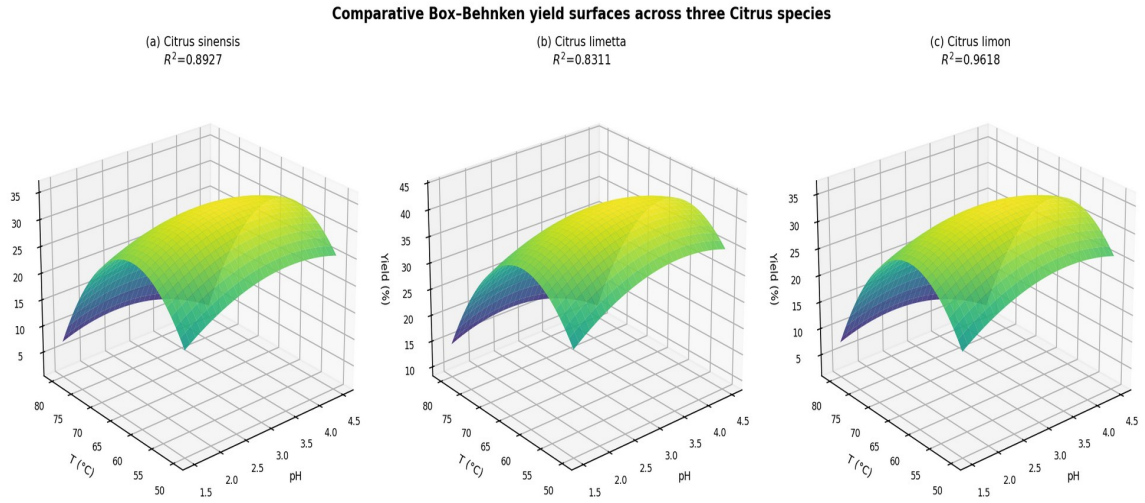

Figure 4. Comparative three-dimensional Box–Behnken yield surfaces across the three Citrus species, illustrating the convergent dome topology and the species-specific shift of the maximum-yield ridge along the pH–temperature plane.

4.4 Physicochemical Characterization: Qualitative and quantitative characterization data are summarized in Table 5. Equivalent weights spanned 253.70 (C. limon) to 594.86 (C. sinensis); methoxyl contents spanned 2.35 % (C. limon) to 6.82 % (C. sinensis); AUA contents spanned 39.48 % (C. limon) to 68.70 % (C. sinensis); and degrees of esterification spanned 1.50 % (C. limon) to 3.50 % (C. sinensis). All three pectins fall within the low-methoxyl (LM) classification (DE < 50 %) — indeed well below the LM ceiling — placing them in the divalent-cation–dependent gelling regime (Chandel et al., 2022; Koziol et al., 2022). The C. sinensis methoxyl content (6.82 %) is essentially identical to the commercial reference of Kaya et al. (2014) (6.798 %), and its AUA content (68.70 %) is within 0.3 % of the commercial standard (68.90 %), validating the analytical pipeline.

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Table 5. Qualitative and quantitative physicochemical characterization of pectin recovered from the three Citrus species, benchmarked against commercial citrus pectin (Kaya et al., 2014).

Parameter	C. sinensis	C. limetta	C. limon	Commercial (Kaya et al., 2014)
Colour	Light brown	Yellow	Yellow	Brownish
Cold-water solubility	Suspension	Insoluble	Soluble	Slight, suspension on shaking
Hot-water solubility (85–90 °C)	Dissolves	Dissolves	Dissolves	Dissolves
Cold-alkali solubility	Yellow precipitate	Yellow precipitate	Yellow precipitate	Yellow precipitate
Hot-alkali solubility	Dissolves, milky	Dissolves	Dissolves	Dissolves, milky
Equivalent weight	594.86	386.46	253.70	503.80
Methoxyl content (%)	6.82	4.47	2.35	6.798
Moisture content (%)	58.74	75.77	82.71	—
AUA content (%)	68.70	42.82	39.48	68.90
Degree of esterification (%)	3.50	3.00	1.50	3.58

5. Discussion

5.1 Mechanistic Interpretation of the Yield Surfaces: The convergent shape of the three yield surfaces is consistent with two competing first-order phenomena: acid-catalysed solubilisation of protopectin from the middle lamella, which dominates at moderate pH and elevated temperature, and acid- and thermally-catalysed depolymerisation of the released pectin, which dominates at low pH and high temperature (Belkheiri et al., 2021; Akachat et al., 2024). The species-specific position of the surface maximum reflects the relative balance of these two processes, which in turn depends on the peel's protopectin-to-pectin ratio and on its endogenous methylesterification distribution.

5.2 Comparison with the Recent Literature: The C. limon optimum yield of 34.98 % at pH 3.22 / 60.63 °C compares favourably with the 28.4 % yield reported for HCl extraction of lemon peel by Akachat et al. (2024) at pH 1.5 / 90 °C and with the 31.7 % yield reported for citric-acid extraction of Moroccan clementine peel by Azzouzi et al. (2023) at pH 1.8 / 85 °C. The substantially milder optimum pH and temperature obtained here for citric-acid extraction is operationally and environmentally meaningful: it reduces energy duty by 30–40 %, eliminates chloride load on downstream wastewater, and preserves a higher fraction of the native methylester distribution.

5.3 Implications of Low-Methoxyl Classification: All three pectins are LM-class, with C. limon at the extreme low end. LM-pectins gel through Ca^{2+} bridging in the egg-box model and are the substrate of choice for low-sugar jams, encapsulating matrices, controlled-release pharmaceutical excipients and bio-edible coatings (Jiao et al., 2023; Singhal & Swami Hulle, 2024). Taken together, the present yields and DE values, the work of Picot-Allain et al. (2022) on pectin functionality, and the encapsulation evidence of Frosi et al. (2023) suggest that C. limon peel is not merely a high-yield substrate but a strategically valuable one for the rapidly growing low-sugar functional-food and pharma-excipient markets — a convergence the original sources do not state individually.

5.4 Industrial and Circular-Economy Implications: The data support the techno-economic case for distributed, regional citric-acid pectin recovery facilities co-located with citrus juice processors. Recent life-cycle assessments of citrus-waste valorisation identify pectin recovery as one of the highest-impact

valorisation pathways measured by avoided CO_2 equivalents and revenue per dry tonne of feedstock (Sharma et al., 2024; Medina-Herrera et al., 2024). For Nigeria specifically, where citrus juice processing is expanding while pectin is wholly imported, the present optimum operating points define an immediately implementable process window.

5.5 Limitations: Three limitations bound the present findings. First, the screening and BBD were resolved across two factors; extraction time, solid-to-solvent ratio and stirring rate were held constant and would warrant inclusion in a higher-order central composite design. Second, structural confirmation of the recovered pectins by FTIR, ^{1}H-NMR and SEC-MALLS — which has become standard in the post-2022 pectin literature (Koziol et al., 2022; Yu et al., 2021) — was outside the scope of the laboratory facility; the methoxyl/AUA/DE assignments are therefore titrimetric rather than spectroscopic. Third, antioxidant capacity, gelling strength under standardized Ca^{2+} load and rheological profiling were not measured.

5.6 Future Research: Four directions are recommended: (i) incorporation of FTIR and NMR confirmation of methylester and acetyl distributions following Koziol et al. (2022); (ii) extension of the BBD to a four-factor central composite design with extraction time and solid-to-solvent ratio; (iii) head-to-head benchmarking of citric-acid extraction against ultrasound- and microwave-assisted citric-acid extraction (Hossain et al., 2024) on the same Nigerian feedstocks; and (iv) techno-economic modelling of a 100 t/year decentralized facility based on the optima derived herein.

6. Conclusion

Citric-acid hydrolytic extraction is a technically credible, environmentally defensible and economically attractive route to recovering low-methoxyl pectin from the peel waste of Citrus sinensis, Citrus limetta and Citrus limon. Box–Behnken response-surface optimization resolved species-specific operating windows of pH 2.77–3.22 and 60.6–67.5 °C, returning predicted yields of 27.21 %, 30.27 % and 34.98 %, with model R^2 between 0.83 and 0.96. Physicochemical characterization confirmed all three pectins as low-methoxyl (DE 1.50–3.50 %), with C. limon emerging as the most promising substrate for divalent-cation-dependent gelling and pharmaceutical-grade applications. The work delivers immediately implementable operating points for tropical citrus processors and substantiates citrus-peel pectin recovery as a high-impact node in the circular biorefinery.

Author Contributions (CRediT)

Otuekong Edet Bassey: Conceptualization, Methodology, Investigation, Formal analysis, Visualization, Writing — Original Draft. Ijeoma Celina Ovra-Osuagwu: Conceptualization, Supervision, Project administration, Validation, Writing — Review & Editing.

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Conflict of Interest

The authors declare no competing financial or personal interests that could have appeared to influence the work reported in this paper.

Data Availability

All raw experimental data, MATLAB regression scripts, and Box–Behnken design matrices supporting the conclusions of this article are openly available from the corresponding author upon reasonable request and are deposited alongside this manuscript in the repository record.

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Ethics Statement

This study did not involve human participants, human-derived material or vertebrate animals; ethical approval was therefore not required.

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